

Four Ideas We Tested -- and Why We Stopped

A research-direction discussion

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Section 1

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This deck is for a research-direction discussion. It does not present a new method. Detailed evidence remains in the source branches and in `docs/saq_research_direction_registry.md`.

1. What I Want To Settle Today

Over the last round, we tested four plausible ways to improve SAQ. Each idea had a reasonable motivation. None produced a result strong enough to become the next paper direction.

I would like us to agree on two things:

- which parts are worth keeping as evidence or baselines; and
- what an idea must show before we open another implementation branch.

The useful outcome of this work is a narrower search space, not a hidden win that needs one more parameter sweep.

2. A Short Reminder: What SAQ Is Doing

SAQ stores a compressed version of each database vector. That makes it cheaper to compare a query with many candidates, but the compressed distance is only an estimate of the true distance.

Three different questions therefore matter:

- **Does the compressed vector resemble the original vector?**
- **Does the estimated distance rank the right candidates near the cutoff?**
- **Does the whole search system remain fast and affordable?**

Several attempts improved the first question without improving the other two. That distinction explains most of the decisions in this deck.

3. The Short Answer

- **Attempt 1 — change or shorten the PCA representation:** the encouraging result did not survive replication, and the lossy version failed even under a deliberately favorable test.
- **Attempt 2 — train scalar codebooks exactly:** useful as a stronger offline baseline, but better reconstruction did not reliably improve search. The inner exact optimizer is also established prior work.
- **Attempt 3 — judge results by distance quality as well as Recall:** this changed one GIST interpretation, but it changed the measurement, not the method, and did not replicate as a positive result on DEEP.
- **Attempt 4 — allow non-power-of-two cardinalities:** mathematically possible, but the registered exact pipeline exceeded its cost ceiling. A later recovery path never produced a valid scientific result and is now closed.

My recommendation is to retain the lessons and close all four method lines.

4. Attempt 1: Can We Represent The Vector Better?

Principal component analysis, or **PCA**, rotates the vector coordinates so that large sources of variation are easier to identify.

We tested two versions of the same broad idea:

- ① keep all dimensions, but learn the rotation from residual vectors rather than raw vectors; and
- ② keep only 576 of 960 dimensions, then store the norm of the discarded tail as a compact summary.

The first version preserves exact geometry before compression. The second throws information away, so we gave it an unusually favorable early test: an exact retained part and an exact tail norm, before adding quantization error.

4A. Attempt 1A: Learn The Rotation From What Search Actually Stores

The first version kept every dimension. Only the data used to learn the PCA rotation changed.

for each database vector x :

 find its assigned IVF center c

 residual $r = x - c$

learn one full-dimensional PCA rotation from all residuals

apply the same rotation to database vectors, centers, and queries

run the unchanged SAQ planner, encoder, and distance estimator

Small illustration — not an experimental result. Suppose two clusters are far apart horizontally, while points vary mainly vertically inside each cluster. Raw-data PCA puts the between-cluster horizontal direction first. Residual PCA removes that separation and may put the within-cluster vertical direction first.

Because this is a full-dimensional orthogonal rotation, exact Euclidean distances are unchanged before quantization. Only the coordinate order seen by SAQ changes.

4B. Attempt 1B: Keep The Head And Replace The Tail By One Norm

The lossy version retained the first 576 PCA coordinates and summarized the remaining 384 coordinates by one tail norm.

head distance = exact squared distance in retained coordinates

tail estimate = squared query-tail norm + squared database-tail norm

estimated distance = head distance + tail estimate

The missing term is the tail inner product: it tells us whether the two tails point in the same or opposite direction.

Running example. Let both retained heads be zero. If the query tail is $(+1)$ and the database tail is also $(+1)$, the true tail distance is zero, while the norm-only estimate is $1 + 1 = 2$. If the database tail is (-1) , both stored norms are still one, but the true distance is four.

One norm cannot distinguish these cases. Rankings can therefore change before any quantization error is added.

5. Attempt 1: What The Evidence Says

On GIST, residual PCA reduced distance-estimation error by about **0.60%** at one accurate stage and **0.33%** with the full code. The fast stage became worse, and the ranking improvement was not reliable.

We then repeated the frozen comparison on CIFAR. The small GIST effect did not repeat: only 4 of 10 seeds favored residual PCA at the accurate stage, the no-rotation control pointed the wrong way, and the fast-stage degradation was supported by the data.

The reduced-dimension version also failed early. Even with exact retained coordinates and exact tail norms, its top-100 agreement was **0.99289**, below native SAQ at **0.99462**. The missing tail inner product—not numerical precision—caused the remaining error.

So we stopped before building a projected index. This does not prove that PCA is universally optimal. It says these two replacements did not earn further systems work.

6. Attempt 2: Does Exact Training Help Search?

A scalar codebook replaces many numeric values with a small set of representatives. Standard Lloyd training is iterative and may settle at a local solution. Exact one-dimensional dynamic programming can find the best partition for the chosen histogram.

That sounds attractive, but it leaves two separate questions:

- Does exact histogram training reduce reconstruction error?
- If it does, are the nearest-neighbor rankings actually better?

The first question is about fitting compressed values. The second is the one the search system ultimately cares about.

6A. Attempt 2: Exact Scalar Training, Step By Step

First sort the values of one coordinate. In one dimension, every optimal cluster is a consecutive interval in that order.

$\text{cost}(a,b)$ = error from replacing sorted values $a\dots b$ by their mean

$\text{DP}[k,m]$ = best cost for the first m values using k levels

$\text{DP}[k,m]$ = min over split j of $\text{DP}[k-1,j-1] + \text{cost}(j,m)$

recover intervals and use each interval mean as a codebook value

repeat for every allowed bit width; allocate the fixed bit budget

encode each value with its nearest learned codebook value

Running example. For values $\{0, 1, 9, 10\}$ and two codebook values, the best split is $\{0, 1\} \mid \{9, 10\}$. The representatives are 0.5 and 9.5. The dynamic program checks every legal split systematically instead of depending on an iterative trainer's starting point.

This guarantees the smallest reconstruction error for the chosen histogram, not the best ranking for future queries.

7. Attempt 2: A Better Fit Was Not A Reliable Search Win

There were genuine positive cases. On the audio dataset, one setting reduced mean squared reconstruction error by about **15.5%** and raised Recall at 100 from **0.95275** to **0.95675**.

But the relationship did not hold consistently:

- increasing histogram resolution kept improving reconstruction while Recall stopped improving monotonically;
- on DEEP at 4 bits, one setting cut raw reconstruction error by about 20% but made Recall at 100 worse; and
- across CIFAR and DEEP, the sign and size of the Recall change depended on the bit budget and allocation objective.

There is also a novelty limit. Exact one-dimensional scalar clustering was already developed by Wu (1991) and later accelerated by Grønlund et al. (2017); the outer bit allocation is a classical discrete allocation problem. We keep this implementation as a demanding offline baseline, not as our method.

8. Attempt 3: Were We Using The Wrong Quality Measure?

Recall at 100 asks whether we returned the same item identifiers as the exact top 100. A distance-quality score asks a softer question: how much farther are the returned items than the exact ones?

This matters near a crowded boundary. Replacing one almost-equally-close item with another hurts Recall, even when the geometric loss is tiny.

We therefore replayed two frozen SAQ plans and reported both measures. Nothing about the encoder, index, or search procedure changed.

8A. Attempt 3: Re-score The Same Returned Items

Attempt 3 did not train a new index. It changed only the offline evaluation.

```
run unchanged search and save its top-k returned identifiers
recompute true distances to those returned items
sort those k distances from smallest to largest
compare position i with the true i-th-nearest distance
average the distance ratios, then invert the result
report this score together with Recall and query speed
```

Running example with $k = 2$. Exact distances are $[1.0, 2.0]$, while the returned items have true distances $[1.0, 2.2]$:

$$\text{distance quality} = \frac{2}{1.0/1.0 + 2.2/2.0} = 0.952.$$

If the second identifier differs, Recall is only $1/2$, although the replacement is just 0.2 farther away. The two metrics answer different questions; neither changes what search returned.

9. Attempt 3: One Decision Changed, But No Method Emerged

At the historical GIST reference point, the alternative plan could not reach the default plan's Recall target. Under the distance-quality measure, however, a directly measured point was slightly better in quality and **1.078 times** faster.

That is worth reporting, but it is not uniform dominance:

- **39.6%** of paired GIST queries became worse;
- the positive conclusion did not carry to the two DEEP controls; and
- the distance-quality measure comes from prior work.

The correct conclusion is narrow: one previous rejection was sensitive to the metric. Future evaluations should show both Recall and geometric quality. We did not discover a new encoding or search mechanism.

10. Attempt 4: Can We Use A Fixed Bit Budget More Flexibly?

Consider two factors that must share a word with at most 16 joint states.

- Restricting each factor to a power of two permits **4 by 4**, using all 16 states.
- Allowing arbitrary positive integers also permits **3 by 5**, using 15 states.

For uneven data, the second shape can fit better while using the same stored word. A synthetic example confirms that this possibility is real.

But a larger set of mathematical choices is not yet a database method. We still have to construct the codebooks at acceptable cost and then show better search quality at comparable speed and memory.

10A. Attempt 4: Exact Curves, Flexible Products, Fixed Words

One coordinate cannot expose this opportunity: with capacity 16, its best choice is simply all 16 levels. Two coordinates are the smallest case where the same word can be divided differently.

```
for each coordinate i:
    compute the globally optimal error  $E_i(K)$ , for  $K=1\dots 256$ 
for each adjacent pair and word capacity S:
    choose  $K_1, K_2$  with  $K_1 * K_2 \leq S$ 
    minimize  $E_1(K_1) + E_2(K_2)$ 
encode  $(z_1, z_2)$  as  $\text{address} = z_1 + K_1 * z_2$ 
```

Running example. Coordinate one has values $\{-1, 0, 1\}$ and coordinate two has $\{-2, -1, 0, 1, 2\}$. With a 4-bit word, $3 \times 5 = 15$ represents every combination exactly. Power-of-two counts force a choice such as 4×4 , so two values on the second coordinate must share a representative.

This proves a representational opportunity, not a search improvement.

11. Attempt 4: The Cost Gate Ended The Study

The registered synthetic implementation passed its correctness checks. The next question was deliberately mundane: can we afford the exact construction and evidence pipeline before reading benchmark data?

The run used **9.67 CPU-hours** and completed only 61 of 128 scalar coordinates. The frozen rule projected two datasets plus a 25% margin, so this partial prefix already implied a lower bound of **24.17025 CPU-hours**, just above the **24-hour** ceiling.

About **4.64 hours** were exact scalar construction and **5.03 hours** were canonical evidence serialization. Allocation, block-vector comparison, and encoding had not started. According to the rule fixed in advance, that was a stop. This is a result about the registered construction-and-evidence pipeline, not evidence that arbitrary cardinalities help or hurt search.

A later attempt to rebuild a cleaner execution path never reached a valid scientific outcome. Its final invocation recorded that it started, but did not produce a trustworthy completion record. We therefore classify the runtime as unknown and have stopped further artifact recovery.

There is no natural-data result, no search result, and no performance claim for this direction.

12. What These Four Attempts Have In Common

Each attempt improved or clarified something local:

- a slightly better distance estimate on one dataset;
- a lower reconstruction objective;
- a more informative evaluation metric; or
- a larger mathematical feasible set.

The local improvement did not survive the next question: replication, ranking, novelty, or cost.

This is the main lesson I would carry forward. We should test the shortest link between an attractive local objective and the final search claim before we build the surrounding machinery.

13. What Is Still Useful

The work was not wasted. We now have:

- SAQ as a well-understood strong baseline;
- exact scalar training as a tougher offline comparison;
- a reason to report both exact-identifier Recall and distance quality;
- examples showing why reconstruction error alone is a poor search proxy; and
- a practical precedent for stopping expensive construction before touching benchmark queries.

These belong in the evaluation design of the next project. They do not need to be promoted into contributions of their own.

14. My Recommendation

Close the four method lines and select the next problem from primary work, before writing more implementation code.

For a new direction, I suggest four entry conditions:

- ① the limitation appears on at least two strong, independent baselines;
- ② the closest published work leaves a specific gap that is not filled by a direct combination of known techniques;
- ③ one cheap, frozen test can reject the idea early; and
- ④ construction, memory, and query cost are part of the acceptance rule from the beginning.

The decision I need from this meeting is whether we agree with that closure and selection rule.

Section 2

Short Technical Appendix

Short Technical Appendix

The main presentation ends here. The remaining slides contain the numbers and source locations most likely to be needed during discussion. Full protocols, implementation details, commit chains, and artifact manifests remain in the registered source documents.

A1. Terms Used In The Talk

- **SAQ:** the compressed-vector baseline studied in this project.
- **PCA:** principal component analysis, a rotation that orders directions by variance.
- **Residual:** the difference between a vector and its assigned cluster center.
- **Recall at 100:** the fraction of exact top-100 neighbor identifiers that appear in the returned top 100.
- **Distance quality, written as 1/Ratio:** how close the returned distances are to the exact top-100 distances; closer to 1 is better.
- **QPS:** queries per second; higher is faster.
- **CPU-hour:** one processor core working for one hour.

A2. Attempt 1 Evidence Boundary

Full-dimensional replacement

- GIST residual-PCA change: accurate-stage error -0.602% , full-code error -0.331% , but fast-stage error worsened.
- CIFAR replication: fast $+0.0317\%$ worse, accurate $+0.00463\%$ worse, full -0.0235% ; confidence and seed requirements failed.
- Decision: close the one-dataset estimator effect.

Lossy 960-to-576 projection

- Favorable projected oracle top-100 agreement: 0.992890625 .
- Native full-dimensional SAQ: 0.994617188 .
- Decision: fail the early ranking gate; projected index construction was not authorized.

A3. Attempt 2 Evidence Boundary

Representative results:

- Audio, 4 bits: reconstruction error per value fell from 280,023 to 236,478; Recall at 100 rose from 0.95275 to 0.95675.
- DEEP, 4-bit rank-boundary setting: raw reconstruction-error ratio was 0.8016, while Recall at 100 changed by -0.00215 .

The result supports two limited claims: exact-histogram training is a useful offline baseline, and lower reconstruction error does not guarantee better nearest-neighbor ranking.

It does not support a new scalar-quantization method claim. Exact one-dimensional dynamic programming is prior work (Wu, 1991; Grønlund et al., 2017), and the outer allocation is a classical discrete rate-allocation problem. White and Singal (2026) is adjacent work: for inner-product accuracy, it weights coordinate errors by how strongly future inputs use those coordinates. That supports our lesson that raw reconstruction error is not the whole downstream objective, but it does not optimize Recall or solve our shared-codebook bit allocation.

A4. Attempt 3 Evidence Boundary

Frozen GIST reference:

- default plan at 200 probes: distance quality 0.999985301, QPS 9264.9;
- alternative plan at 280 probes: distance quality 0.999987869, QPS 9986.9;
- measured speed ratio: 1.07793.

Query-paired result: 39.6% worse, 23.0% equal, and 37.4% better. Both DEEP controls remained below their default distance-quality targets.

Decision: keep this as metric-sensitivity evidence. Do not claim per-query dominance, cross-dataset replication, or a new mechanism.

A5. Attempt 4 Boundary

The synthetic 16-state example showed a real feasible-set difference:

- arbitrary (3,5) cardinalities: zero distortion;
- power-of-two (4,4) cardinalities: 0.1 distortion per vector.

This was instrument validation only. No benchmark data or search evaluation was involved.

The partial run consumed 9.668 CPU-hours for 61 of 128 scalar coordinates. The frozen two-dataset-plus-margin rule converted that completed prefix into a strict lower bound of 24.17 CPU-hours against a 24.0 CPU-hour ceiling. Exact scalar construction used 4.64 hours and canonical evidence serialization used 5.03; all later phases remained unexecuted. The registered decision was `NO_GO_EXACT_SOLVER_COST` for that full frozen pipeline.

The later V2 execution path produced no valid terminal record. The audited project decision is to stop further artifact recovery; the runtime remains unknown rather than being reclassified as a scientific failure.

A6. What We Can And Cannot Say

Supported:

- the tested PCA replacement did not replicate;
- the registered norm-only projection failed its favorable early ranking test;
- exact scalar training improved offline fit more reliably than search;
- one GIST comparison changed under a distance-based quality measure; and
- Attempt 4 stopped on registered pipeline cost, followed by project closure without a scientific V2 result.

Not supported:

- PCA is universally optimal;
- exact scalar training never helps search;
- distance quality makes the alternative SAQ plan generally better;
- arbitrary cardinalities cannot help on real data; or
- any of these attempts produced a new method or moved the quality-speed frontier against the strongest baseline.

A7. Authoritative Source Snapshots

- Attempt 1A: `saq-transform-analysis@3d94840`.
- Attempt 1B: `saq-lossy-projection-analysis@051ec6a`.
- Attempt 2: `vectordb@f51b487`; prior-work boundary at `saq-caq-one-shell-repair@433e8ea`.
- Attempt 3: `saq-ratio-metric-analysis@146dc16`.
- Attempt 4 cost stop: `saq-arbitrary-cardinality-analysis@f1b464b`.
- Attempt 4 project closure: `saq-arbitrary-cardinality-feasibility-v2@3577edd`.

A8. Authoritative Evidence Documents

- Attempt 1A:
docs/saq_transform_phase1b_external_replication_evidence_2026_07_10.md.
- Attempt 1B: docs/saq_lossy_projection_lp0_gate_a_evidence_2026_07_11.md.
- Attempt 2: reports/scalar_training_exact_hist_audit_2026_06_30/README.md and docs/saq_limitation_transfer_memo_2026_07_02.md.
- Attempt 3: docs/saq_attempt3_a3_2_a3_3_decision_2026_07_13.md.
- Attempt 4 cost stop:
docs/saq_attempt4_a4_1s_cost_projection_review_2026_07_13.md.
- Attempt 4 project closure:
docs/saq_a4_v2_project_stop_no_further_artifact_recovery_2026_07_18.md.
- Cross-attempt reporting state: docs/saq_research_direction_registry.md.

No further research or execution step is currently authorized for these four method lines.

A9. Selected References For Attempt 2

- X. Wu. “Optimal Quantization by Matrix Searching.” *Journal of Algorithms*, 1991.
<https://www.sciencedirect.com/science/article/pii/0196677491900392>
- A. Grønlund, K. G. Larsen, A. Mathiasen, J. S. Nielsen, S. Schneider, and M. Song. “Fast Exact k-Means, k-Medians and Bregman Divergence Clustering in 1D.” 2017.
<https://arxiv.org/abs/1701.07204>
- N. White and K. Singal. “Inner Product Aware Quantization: Provably Fast, Accurate, and Adaptive Algorithms.” arXiv:2606.00289v1, 2026.
<https://arxiv.org/abs/2606.00289v1>